

MECHANISMS OF DRUG TRANSPORT

Mechanisms of drug transport

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- Two main mechanisms of drug transport
 - ✓ **Transcellular absorption:** is the process of drug movement across a cell.
 - ✓ **Paracellular absorption:** Some polar molecules may not be able to traverse the cell membrane but, instead, go through gaps or tight junctions between cells, a process known as **Paracellular absorption**
- Some drugs are probably absorbed by a mixed mechanism involving one or more processes.

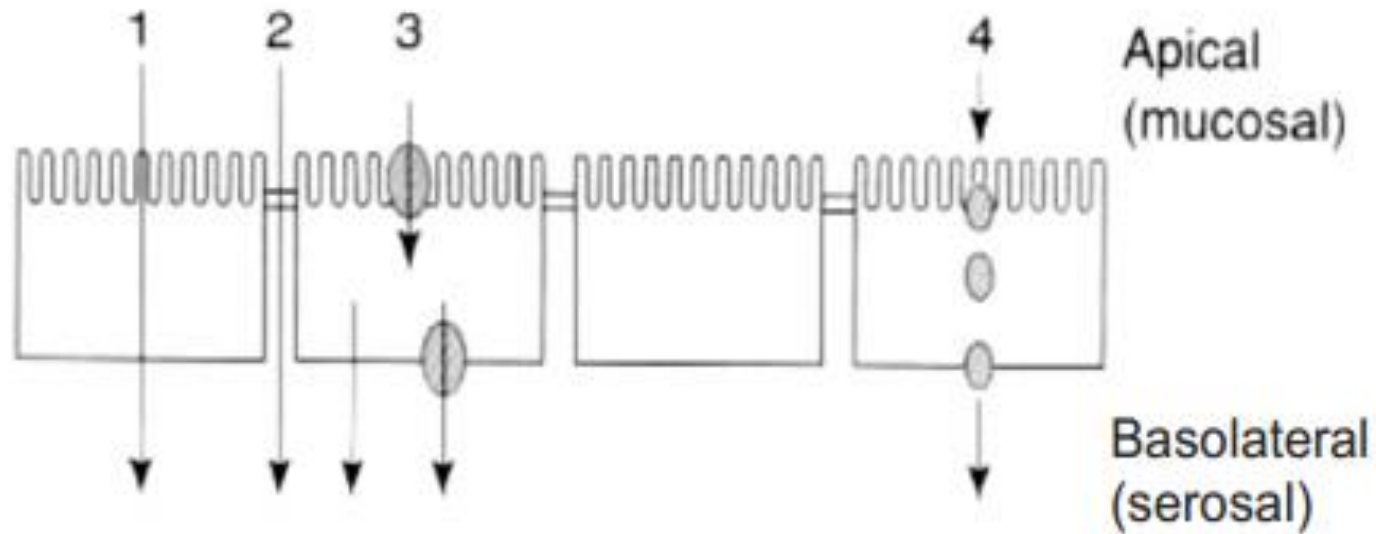
Mechanisms...

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- Transcellular mechanisms:
 - ***Simple passive diffusion***
 - ***Carrier-mediated transport***
 - Active transport
 - Facilitated diffusion
 - ***Endocytosis***
 - ***Miscellaneous:*** Ion-pair Formation, convective/pore transport

Mechanisms...

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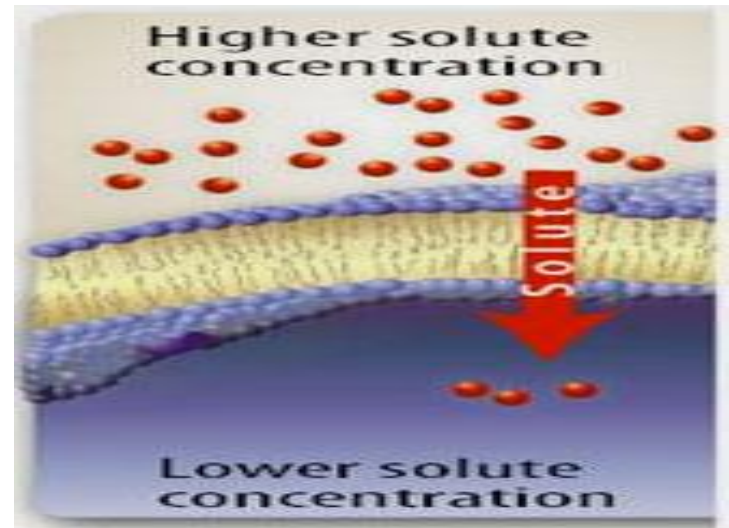
1 – Transcellular 3 – Carrier mediated
2 – Paracellular 4 – Transcytosis

Fig. Summary of intestinal epithelial transport mechanisms.

Passive diffusion

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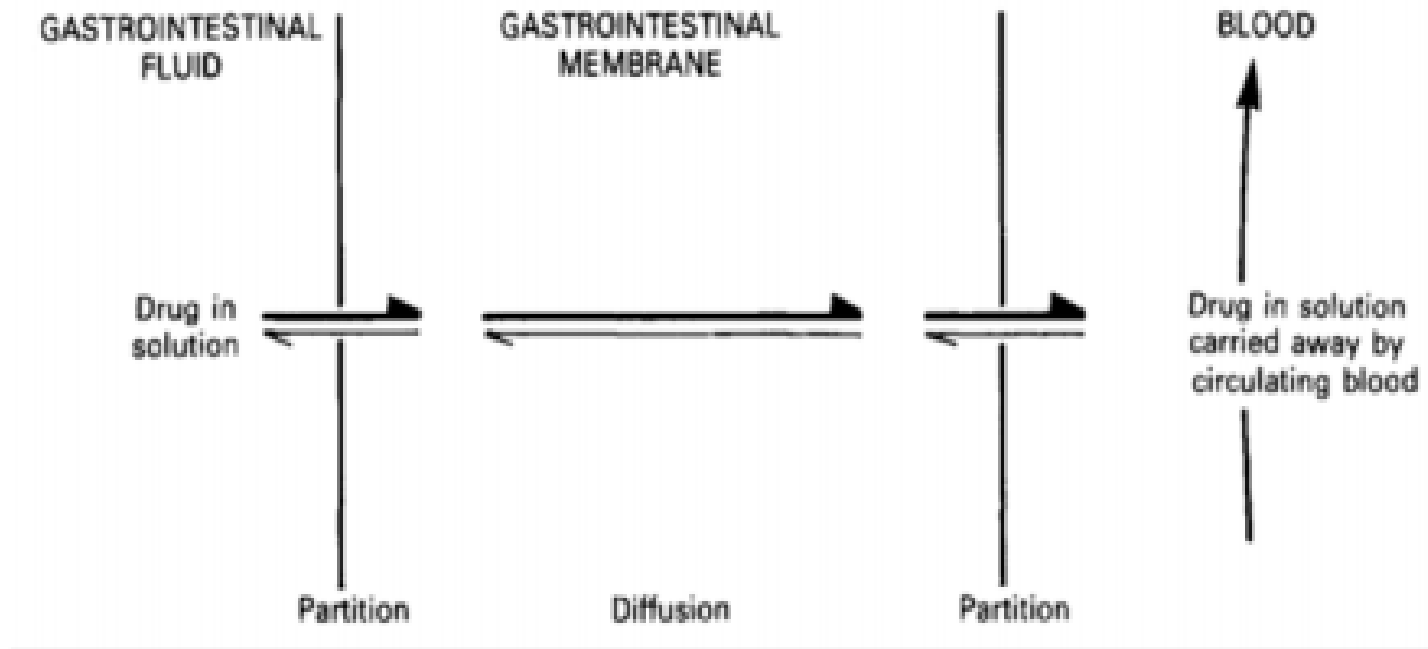
- The process by which molecules spontaneously diffuse from a region of higher concentration to a region of lower concentration.
- Preferred for relatively small lipophilic molecules
- The lipid cell membrane is not a barrier for the diffusion and absorption of such drug.



Passive diffusion...

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- Drug molecules move forward and back across a membrane.



Passive diffusion...

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- If the two sides have the same drug concentration, forward moving drug molecules are balanced by molecules moving back, resulting in **no** net transfer of drug.
- When one side is higher in drug concentration, at any given time, the number of forward-moving drug molecules will be higher than the number of backward-moving molecules;
 - the net result will be a transfer of molecules to the alternate side

Passive diffusion...

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- Passive diffusion is the major absorption process for most drugs.
- Passive diffusion indicates that the transfer of a compound through a membrane may be described by:
 - physicochemical laws and
 - the properties of the membrane.

Passive diffusion...

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- The driving force for diffusion across the membrane is the **concentration gradient** of the compound across that membrane.
- Passive diffusion is best expressed by **Fick's first law of diffusion**
 - ❖ *States that the drug molecules diffuse from a region of higher concentration to lower concentration until equilibrium is attained and that the rate of diffusion is directly proportional to the concentration gradient across the membrane.*

Passive diffusion...

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$$\frac{dQ}{dt} = \frac{DAK}{h} (C_{GI} - C_p)$$

Where:

- dQ/dt - rate of diffusion
- D - diffusion coefficient
- A - surface area of membrane
- K - partition coefficient
- h - thickness of the membrane
- $(C_{GI} - C_p)$ - difference in the conc. of drug in the GI fluids and the plasma.

Passive diffusion...

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- Since the volume into which the drug may distribute from the blood is large compared to the gut fluid volume and since the rapid circulation of blood through the GIT continually moves absorbed drug away from the site of absorption, $C_{GI} \gg C_P$
 - Drugs are usually given in milligram doses, whereas plasma concentrations are often in the $\mu\text{g/ml}$ or ng/ml range.
- Because D , A , K , and h are constants under usual conditions for absorption, a combined constant P or *permeability coefficient* may be defined.

Passive diffusion...

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- As a result the equation can further be simplified:

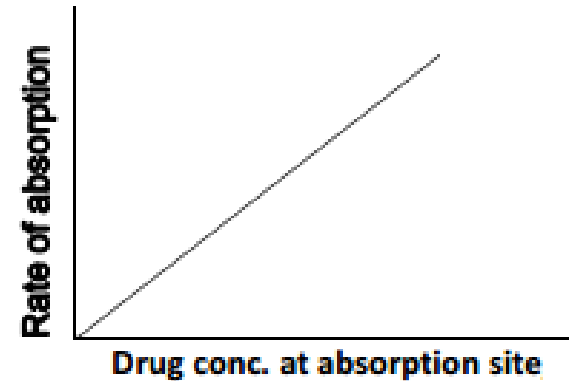
$$\frac{dQ}{dt} = P(C_{GI})$$

- It is an expression for a first-order process.
 - Passive drug absorption generally adheres to first order kinetics.
- The rate of absorption should increase directly with an increase in drug concentration in the GI fluids.

Passive diffusion...

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- The slope of this line represents the constant (P) for the absorption



- For drugs that act as weak electrolytes, such as weak acids and bases, the extent of ionization influences the rate of drug transport.

GI membrane is more permeable to unionized form (greater lipid solubility)

Carrier-mediated transport

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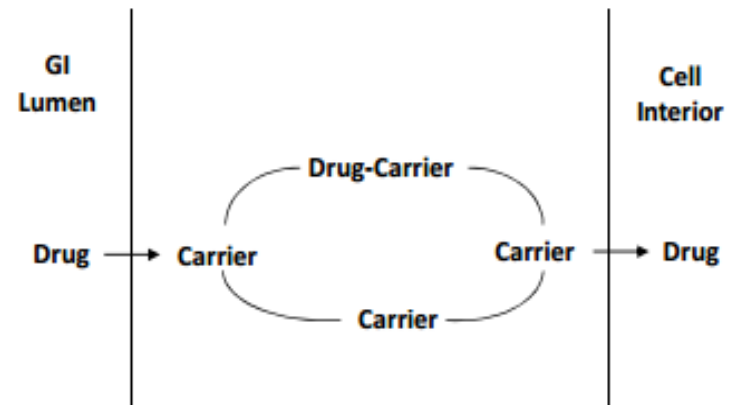
- Most drugs are absorbed from GIT via passive diffusion.
- Many essential nutrients are water-soluble and they have low oil/water partition coefficients, which would suggest poor absorption from the GIT.
- To ensure adequate uptake of these materials from food, the intestine has developed specialized absorption mechanisms that depend on membrane participation and require the compound to have a specific chemical structure.

➤ ***Carrier mediated transport***

Carrier-mediated transport...

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- Few *lipid insoluble drugs* and many *nutrients* are absorbed by carrier-mediated transport
 - Carrier binds drug and transports it across membrane
- Explained by shuttling process across epithelial membrane
 - Drug forms complex with carrier
 - Drug-carrier complex moves across membrane
 - Drug liberated on other side
 - Free carrier returns to initial position in cell membrane adjacent to GI lumen



Carrier-mediated transport...

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- Carrier-mediated transport
 - Active transport
 - Facilitated transport

Active transport

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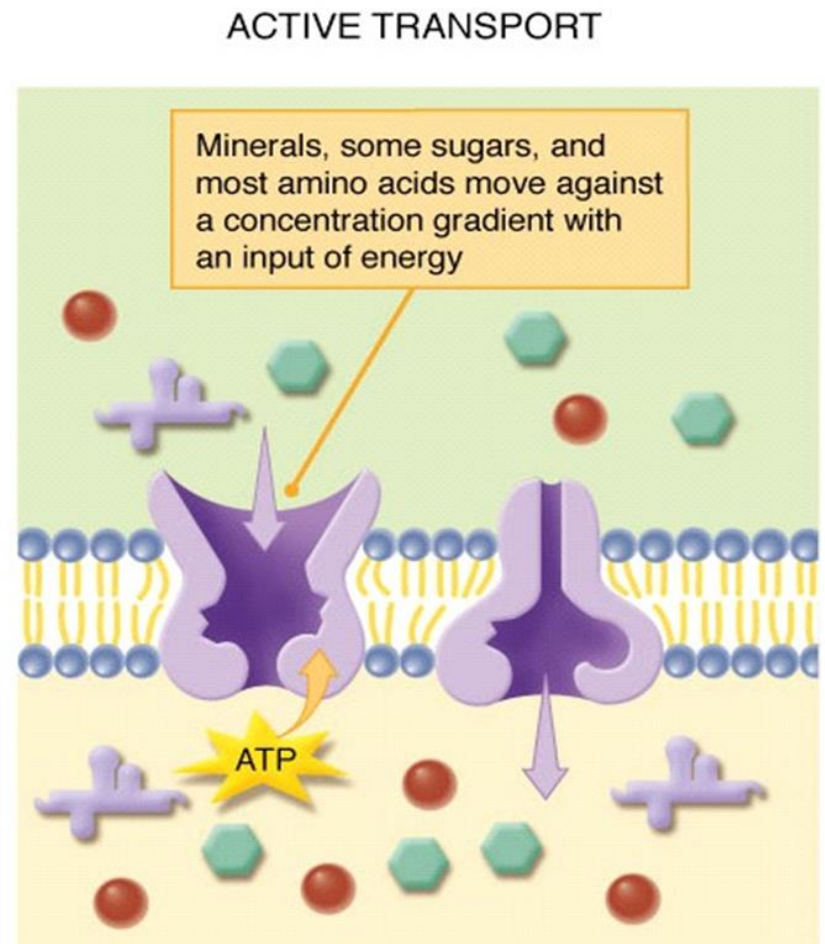
- Carrier-mediated transmembrane process that plays an important role in the gastrointestinal absorption and in renal and biliary secretion of many drugs and metabolites.
- It is characterized by the transport of drug **uphill** against a concentration gradient.
 - *this is an energy-consuming system.*
- Developed for nutrients and chemicals essential to life

Active transport...

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□ Large number of active transport systems in SI

- Peptide T, sugar T
bile acid T, amino acid T,
vitamin T...
- Each carrier system is conc.
in specific segment of GIT and
has its own substrate specificity



Active transport...

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- Drug structurally resembling natural substance (“false” nutrients) that is actively transported is likely to be transported by same carrier
 - Penicillins, cephalosporins, and ACE inhibitors rely on *peptide transporters* for efficient absorption
 - Nucleosides and their analogues for antiviral and anticancer drugs depend on *nucleoside transporters*
 - ✓ 5-fluorouracil is transported by pyrimidine transport system
 - L-dopa and α-methyldopa are transported by *amino acid transporters*

Active transport...

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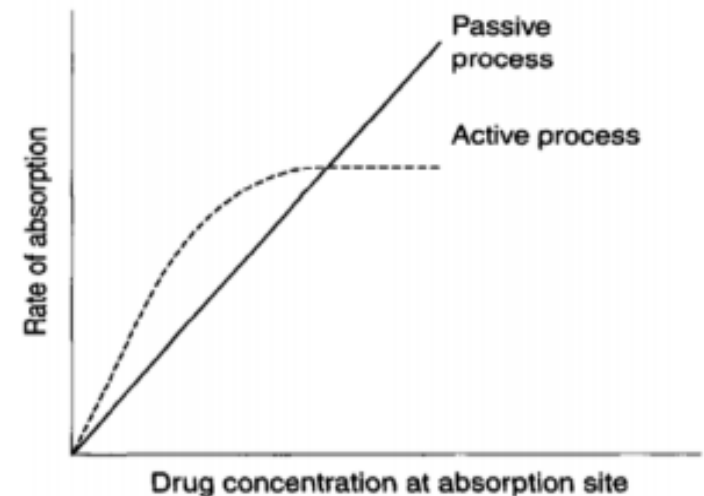
Table: Intestine transporters and examples of drugs transported

Transporter	Examples	
Amino acid transporter	Gabapentin	D-Cycloserine
	Methyldopa	Baclofen
	L-dopa	
Oligopeptide transporter	Cefadroxil	Cephadrine
	Cefixime	Ceftibuten
	Cephalexin	Captopril
	Lisinopril	Thrombin inhibitor
Phosphate transporter	Fostomycin	Foscarnet
Nucleoside transporter	5-fluorouracil	
Glucose transporter	p-Nitrophenyl- -D-glucopyranoside	
Monocarboxylic acid transporter	Salicylic acid	Benzoic acid
	Pravastatin	

Active transport...

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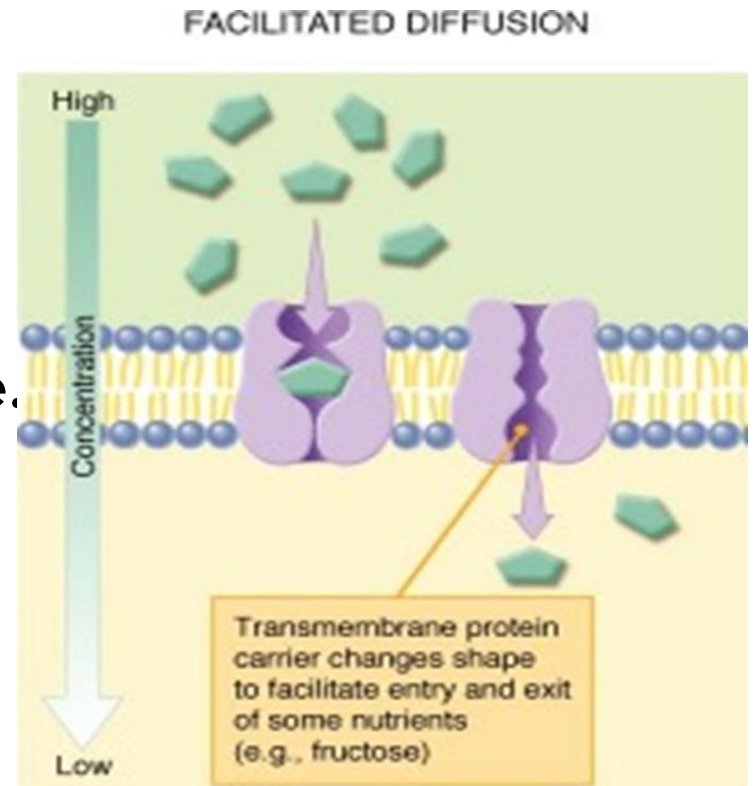
- When a drug is absorbed by a carrier-mediated active process,
 - ❖ Rate of absorption $\propto$ conc. of absorbable species only at low conc.
 - ❖ Becomes saturated at higher conc.
 - ❖ Capacity limited process
 - ❖ Further increase in conc. will not increase rate, i.e., rate of absorption remains constant, or **zero order**.



Facilitated diffusion

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- Solutes are transported **downhill** but at much faster rate than would be anticipated based on molecular size and polarity
- In terms of drug absorption, facilitated diffusion seems to play a very minor role.



Vesicular transport

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- The process of engulfing particles or dissolved materials by the cell.
- Pinocytosis and phagocytosis are forms of vesicular transport that differ by the type of material ingested.
- **Pinocytosis** refers to the engulfment of small solutes or fluid,
- **Phagocytosis** refers to the engulfment of larger particles or macromolecules, generally by macrophages.
- **Endocytosis** and **exocytosis** are the processes of moving specific macromolecules into and out of a cell, respectively.

Vesicular transport...

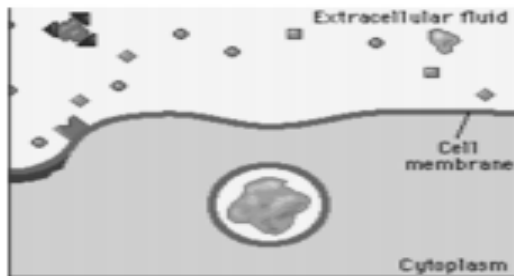
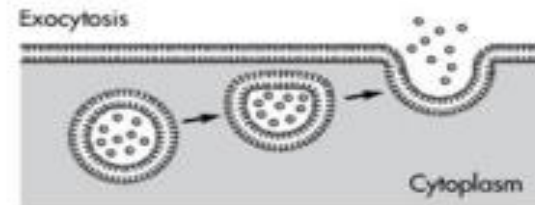
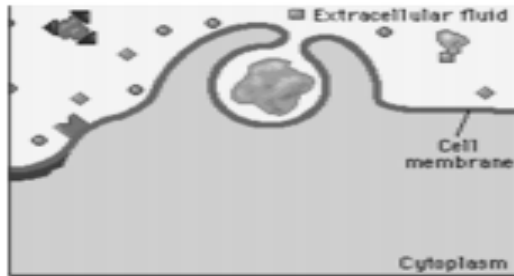
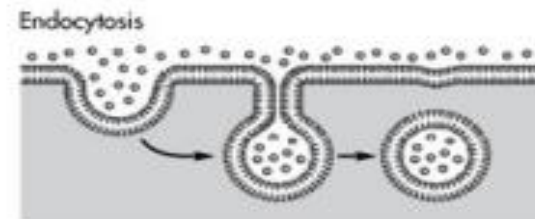
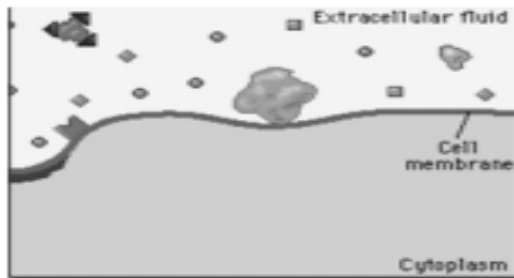
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- During pinocytosis or phagocytosis, the cell membrane invaginates to surround the material and then engulfs the material, incorporating it into the cell.
- The cell membrane containing the material becomes pinched off, forming small intracellular membrane-bound vesicle or vacuole within the cell.
- Material migrate to basolateral surface of cell and exocytosed

Vesicular transport...

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Diagrams showing vesicular transport: *endocytosis* and *exocytosis*



Vesicular transport...

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- Endocytosis is primary mechanism of transport of macromolecules
 - orally administered Sabin Polio vaccine and various large proteins.
- An example of exocytosis is the transport of a protein such as insulin from insulin-producing cells of the pancreas into the extracellular space.

Convective (pore) transport

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- Very small molecules such as water, urea and low molecular weight sugars are able to cross cell membranes rapidly as if there is no barrier for their passage
- Explained by aqueous filled pores or channels assumed to be present in cell membrane
- A certain type of protein called a *transport protein* may form an open channel across the lipid membrane of the cell.

Convective transport...

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- Small molecules including drugs move through the channel by diffusion more rapidly than at other parts of the membrane
- Radius of channel is estimated to be in order of 0.4 nm
- Due to molecular size limitations, *convective transport* is of minor importance in GI absorption of large water soluble drug molecules or ions.

Ion-pair formation

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- Strong electrolyte drugs such as quaternary nitrogen compounds maintain their charge at all physiologic pH values and penetrate membranes poorly.
 - Cannot partition directly into lipoidal membrane
 - Too large to pass through aqueous filled pores in membrane
- When the ionized drug is linked up with an oppositely charged ion, an **ion pair** is formed in which the overall charge of the pair is neutral.

Ion-pair formation ...

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- This neutral drug complex diffuses more easily across the membrane.
- For example, the formation of ion pairs to facilitate drug absorption has been demonstrated for:
 - propranolol, a basic drug that forms ion pair with oleic acid
 - quinine, which forms ion pair with hexylsalicylate

Paracellular pathway

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- Small polar molecules ($K_{o/w} < 1$) may have access to a convoluted route that exists between adjacent epithelial cells.
 - This route is referred to as being *paracellular*.
- The epithelial cells are joined together via closely fitting tight junctions on their apical absorbing surface.
- The intercellular spaces occupy only about **0.01%** of the total surface area of the epithelium.

Paracellular pathway...

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- The tightness of these junctions can vary considerably between different epithelia in the body.
- In general, absorptive epithelia, such as those of the SI, tend to be leakier than other epithelia.
- The paracellular route of absorption is important for the transport of ions such as *calcium* and for the transport of *sugars, amino acids* and *peptides* at concentrations above the capacity of their carriers.

Paracellular pathway...

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- Small hydrophilic and charged drugs that do not distribute into cell membranes cross the GI epithelium via the paracellular pathway.
- The molecular weight cut-off for the paracellular route is usually considered to be 200 Da, although some larger drugs have been shown to be absorbed via this route.